

Firedrake/UFL verification of a structural scalar-field model: spin- $\frac{1}{2}$ FR signatures, hydrogenic radial gates, and solar-system perihelion tests

A verification-ready protocol for testing a single-field structural-medium model, now including selected Firedrake/UFL real-solver bridge runs (v0.3 draft)

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Abstract

“The Standard Model answers what we observe with extraordinary precision. SFT asks a different question: under what structural conditions do the observed stable configurations exist at all? This is a shift from phenomenological description to constraints on existence and stability. It is not presented as competition, but as complementarity.”

Structural Field Theory (SFT) is a monistic structural-medium framework in which observed phenomena are modeled as stable configurations of a single structural field variable S . Effective phase, topological, electromagnetic, and gauge-like descriptions are extracted only under declared domain conditions, rather than introduced as freely tunable independent degrees of freedom. This v0.3 draft keeps the release verification-first: it provides artifacts, schemas, hash manifests, and pre-registered pass/fail gates so independent reviewers can evaluate claims without trusting the authors. The update adds a Firedrake/UFL bridge package with selected real-solver/exploratory audit targets, including robust spin-1/2 FR bend signatures across meshes, hydrogenic radial Coulomb H/T 2s/2p gates, and solar-system perihelion geometry gates for Venus and Mercury. The v0.1.9 Firedrake bridge further adds a paired scalar negative-control and spin/topological-memory test. These bridge runs are not presented as global validation of SFT, nor as derivations of the Standard Model or General Relativity. They demonstrate that selected SFT-facing audit targets can be executed in an external PDE/FEM environment and reported through machine-checkable PASS/FAIL artifacts. Physical confirmation of SFT remains conditional on third-party real-solver outputs at multiple resolutions under the published gates.

Keywords:

Verification protocol; reproducibility; external audit; artifact contract; schema validation; pre-registered thresholds; computational physics; emergent phenomena; Firedrake.

1. Introduction

Many physical quantities currently enter our best models as inputs: coupling constants, particle properties, and medium-like parameters that are measured rather than derived. SFT explores an alternative stance: treat the “vacuum” as a structural medium and treat what we call matter as stable, medium-compatible configurations of a single underlying field.

This paper is deliberately verification-first: we focus on what an external reviewer can run, check, and falsify [1,2].

SFT is not presented as a replacement for the Standard Model, but as a complementary structural-medium hypothesis aimed at explaining why the observed stable configurations exist. The Standard Model remains the reference phenomenology; SFT targets a causal/mechanistic layer beneath it. Nothing in this release disputes the Standard Model’s empirical success; the intent is a reinterpretation at the level of existence/conditions, evaluated only through pre-registered gates and external runs.

- Deliverable: a protocol and tooling that minimizes trust in authors (schemas, manifests, deterministic seeds, pass/fail gates).
- Scope: EM + gravity/PPN bridge + an emergent spin- $\frac{1}{2}$ route; not a full Standard Model replacement [8].
- Status: verification-ready with selected Firedrake/UFL bridge runs; not yet globally confirmed end-to-end without independent third-party solver outputs.

Physics-Facing Audit Targets and Current Status

This release is not a narrative-only protocol: it includes representative physics-facing audit targets distributed as runnable bundles, each tied to explicit PASS/FAIL gates, SHA-256 manifests, and, where applicable, schema-validated artifacts. A compact status table is kept in the main text; the longer bundle inventory is provided in Appendix A.

To avoid overclaiming, every bundle is labeled by evidential role: CI checks verify tooling; REAL-ready modules define contracts for external solver outputs; Firedrake bridge runs are exploratory producer-side outputs generated in an external PDE/FEM environment; and REAL-verified claims remain reserved for independent third-party runs under locked gates.

Table 1. Compact evidential status of representative audit targets.

Audit layer	Representative bundle / artifact	Current status
CI audit pipeline	Double-slit synthetic gate; EOC CI; AtomX; External Review Pack; alpha-like stress test	CI (C): tooling, schemas, manifests, gates
REAL-ready contracts	Alpha-out runner; Venus/perihelion verifiers; spin Appendix S	REAL-ready (P-input): can ingest third-party outputs
Firedrake/UFL bridge	SFT_FIREDRAKE_CORE_REPRO_v0_1_9	Exploratory P-bridge with real-solver outputs
Headline Firedrake results	Scalar recovery/residue control; spin topological memory; radial H/T Coulomb; Venus/Mercury perihelion gates	Exploratory PASS under package gates
Open roadmap branches	Firedrake double-slit; Alpha-Out bridge	Double-slit MIXED/HOLD; Alpha-Out INCOMPLETE

Interpretation.

- CI (C) means CPU-only validation of the audit pipeline (artifacts, gates, schemas, manifests). It is not end-to-end physical confirmation by itself.

- REAL-ready (P-input) means the verifier and artifact contract are in place and can ingest third-party REAL solver outputs.

- REAL-verified (P) would require an independent REAL solver run, published with artifacts + hashes, that passes the declared gates.

Within this testbed, the Firedrake v0.1.9 result should be read as the combination of a negative scalar control and a positive topological-memory gate. The scalar recovery/residue runs disperse localized impulses without crossing the persistent-residue gate, reducing the likelihood that the signal is due to remnant scalar contamination. By contrast, the director/spin bridge retains a robust Finkelstein-Rubinstein memory across mesh resolution, loop resolution, and localization-width scans. Thus, within the stated model and gates, the retained memory is best interpreted as topological/holonomic rather than as a scalar-residue artifact.

Note on α -out.

The packs `SFT\_alpha\_out\_MINI\_bundle\_v0.3-FINAL\_FIXED` and `SFT\_alpha\_out\_MINI\_runner\_ready\_v3\_with\_schema\_checks\_FIXED` implement the same α -out verification logic at different packaging levels; the latter adds explicit schema-checked runner readiness.

2. Minimal framework (what is being tested)

SFT assumes a single structural field variable S (dimensionless in the discrete model), interpreted as the structural/tensional degree of freedom of the medium often called “vacuum.” The scalar representation is operational: effective phase, topological, spin-like, and electromagnetic descriptions are extracted from S -derived artifacts only within declared regimes and domain-of-validity checks. Matter corresponds to stable, localized or topological configurations of S ; dynamics is discrete-first with a controlled continuum limit.

Interpretation (minimal): S encodes the local structural/tensional state of the medium. Stable, medium-compatible configurations of S correspond to what we describe as matter; propagating disturbances correspond to radiation modes. All statements in this paper are operationalized via exported artifacts and pass/fail gates.

2.0 Minimal physics backbone (orientation-only)

This protocol paper is intentionally verification-first; the purpose of this short backbone is to state the minimal physical model being exercised by the runners, without expanding into a full derivation (which remains in the companion technical corpus).

- State variable. $S(x,t)$ is represented as a single structural field variable (dimensionless in the discrete model). Discrete simulations represent S on a lattice for computation; phase, topology, spin-like signatures, and gauge-like mappings are treated as extracted descriptions under declared domain conditions, not as hidden independent degrees of freedom. Physical statements are interpreted under declared domain-of-validity checks and mesh-control diagnostics.
- Minimal equation of motion (orientation). A representative continuum form is:
$$\partial_t^2 S - c^2 \nabla^2 S + \partial V / \partial S = 0$$
 (often written as $\partial_t^2 S - c^2 \nabla^2 S + \alpha V S + \lambda_4 S^3 = 0$ for the minimal quartic potential).
- Energy budget. For non-collapse evolution, the local energy density is tracked as $u = \frac{1}{2}(\partial_t S)^2 + \frac{1}{2}c^2 |\nabla S|^2 + V(S)$, with a corresponding global energy integral used for PASS/FAIL budgets. When collapse-style effective rules are enabled, any energy removed from the propagating field is recorded explicitly in a “collapse ledger” artifact to preserve auditability.
- Discrete manifestation. Although S is continuous in availability, physically relevant “objects” are defined as stable families/attractors that persist under perturbations and survive mesh refinement; particle labels (electron/proton/neutron, etc.) are treated as equivalence classes identified by invariant signatures and preregistered gates, not as a unique micro-geometry.

- Structural collapse vs measurement. Structural collapse denotes an intrinsic regime change of the medium when local propagation/relaxation limits (bounded by c and the medium response) are exceeded; it may occur in free evolution. Measurement contexts (e.g., a screen in a double-slit setup) are modeled as external interventions (apparatus-defined coupling/boundary forcing) that can trigger the same structural collapse locally, with “quantum collapse” reserved for the readout/update layer used to report outcomes.

- We state falsifiable predictions: (I) high- k modified dispersion (stringent tests of Lorentz invariance), (II) sub-mm Yukawa-like deviations in gravity, (III) cosmic birefringence, and (IV) the onset of nonlinear Breit–Wheeler pair production mapped in strong-field parameters (a_0, χ_Y) rather than a single intensity threshold. Together, these proposed tests position SFT as a reproducible, falsifiable program that bridges lattice field simulations with quantum and relativistic observables. These are proposed falsifiable tests; they are not validated in this release.

2.1 Minimal dynamics (presentation-level)

We keep the main text minimal; detailed model choices live in the accompanying corpus.

Continuum reference (for orientation):

$$\mathcal{L}_0 = 1/2 (\partial_\mu S)(\partial^\mu S) - V(S)$$

Discrete runs must demonstrate controlled systematics (energy/continuity budgets, dispersion fits, convergence checks) [6,7].

2.2 Operational notion of existence (natural vs maintained)

A configuration “exists” in SFT if it is stably compatible with its medium, in a way that can be certified over a non-empty region of medium parameters under declared gates.

- Natural existence: stability holds in the ambient medium (no declared upkeep).
- Maintained existence: stability holds only when minimal declared controls (BCs, confinement, fields, filtering, etc.) are applied; ablation removes PASS.

2.3 Exclusive falsifiable predictions

The following predictions distinguish SFT from the Standard Model in experimentally accessible regimes. They are not validated in this release; they constitute the falsifiable program that future independent runs should target.

Table 2. Exclusive falsifiable predictions

Prediction	Key relation / model	How to test (observable)	Notes / typical scale
High-k Lorentz-violation in dispersion	$\omega(k) = c \cdot k \cdot \sqrt{1 + \xi \cdot (a \cdot k)^2} \approx c \cdot k \cdot (1 + \frac{1}{2} \cdot \xi \cdot (a \cdot k)^2)$ for $a \cdot k \ll 1$	Measure phase/group velocity vs k ; fit quadratic correction to $\omega/(c \cdot k)$ as a function of $(a \cdot k)^2$.	a = lattice spacing; $\xi = O(1)$ dimensionless sign. Plot $\omega/(c \cdot k)$ vs $(a \cdot k)^2$ (slope $\approx \xi/2$ at small $a \cdot k$). Assumptions: isotropic lattice; weak-violation regime $a \cdot k \ll 1$; negligible damping. Status: simulation-ready.
Cosmic birefringence (parity violation)	$\Delta C_{\ell}^{\{EB\}} \approx \epsilon_p \cdot C_{\ell}^{\{EE\}}$; rotation angle $\alpha_{\text{rot}} \approx \epsilon_p/2$ (small-angle mixing)	CMB polarization: EB cross-spectrum; estimate α_{rot} from QU maps or EB/EE.	Compare to Λ CDM null EB; control beam and calibration systematics. Assumptions: uniform small rotation; negligible foreground leakage after cleaning. Status: data-ready (method).
Sub-millimeter Yukawa correction to gravity	$V(r) = -G \cdot m_1 \cdot m_2 / r \cdot [1 + \alpha_Y \cdot \exp(-r/\lambda)] / F(r)/F_{\text{Newton}}(r) = 1 + \alpha_Y \cdot (1 + r/\lambda) \cdot \exp(-r/\lambda)$	Torsion balance / micro-cantilevers: residual force vs separation; fit (α_Y, λ) .	$\lambda \lesssim \text{mm}$; $\alpha_Y \geq 0$ in this scenario; include electrostatic/patch controls. Assumptions: point-mass approximation; stable temperature and vibration isolation. Status: lab-ready.
Breit–Wheeler pair creation — Linear (two-photon)	Kinematic threshold in energy: $s = (p_Y + p_{Y'})^2 \geq 4 m_e^2 c^4$ (i.e., $E_Y + E_{Y'} \geq 2 m_e c^2$ in head-on frame)	γ – γ luminosity (collider or engineered photon bath); observe e^+e^- yield vs photon energy and overlap.	No universal intensity threshold; falsifiability via luminosity and spectrum control. Status: design-ready (method-level).
Breit–Wheeler pair creation — Nonlinear (multiphoton)	Onset: $\chi_Y \gtrsim 1$ (or $a_0 \gtrsim 1$ with GeV γ/e probes); rates depend on $\{a_0, \chi_Y, \lambda, \text{pulse shape}\}$	GeV electron/ γ beam colliding with ultra-intense Ti:sapphire PW laser; measure e^+e^- yield vs intensity and probe energy.	Facility scale today: $I \approx 10^{20}–10^{23}$ W/cm ² (ELI/Apollon/CoReLS class). Report SFT as rate maps vs $\{a_0, \chi_Y, \lambda, \gamma\}$; do not quote a single I_{thr} . Status: lab-proposed (facility-dependent).

3. Operational dictionary and audit artifacts

3.1 Compatibility operator and “existence region”

For a target configuration S and medium levers M , SFT defines a compatibility operator $\mathcal{C}[S;M]$ collecting constraints such as continuity, energy ledgers, dispersion control, EM consistency constraints (as implemented), and (when applicable) PPN mapping checks. Existence is certified by scanning or optimizing M to minimize $\|\mathcal{C}\|$ and by exporting an existence region $\mathcal{R}$ where gates PASS. Constraints are implemented as explicit verifiers over exported artifacts, not as hidden fit degrees of freedom.

3.2 Electromagnetism: derived potential and domain of validity (brief)

In this release, electromagnetism is treated as a derived description over the structural field rather than as an additional independent set of degrees of freedom. Specifically, the effective phase/angle variable θ is defined only under explicit coarse-graining assumptions (e.g., narrowband / slowly-varying regimes), and the electromagnetic potential A_μ is introduced as a functional of exported S-artifacts via a fixed Green/operator construction. A_μ is therefore not an independently tunable DOF, although it may be non-local at the potential level.

- Domain notes:
- θ -definition: valid in regimes where phase extraction is stable under declared windows (narrowband / coarse-grain). Outside those regimes, θ is treated as undefined rather than forced.
- Near $R \approx 0$ (core/defect center): EM mapping may require excision/regularization; the excision rule and window are treated as declared inputs and must be reported in SCAN metadata.
- No extra DOF: any apparent freedom in A_μ is fixed by the chosen Green/operator and boundary conditions; changes must be treated as model variants and pre-registered.

3.3 Artifact contract (SCAN → REGION → REPORT)

Any externally verifiable claim must provide a minimal artifact chain:

- SCAN: raw sweep results (CSV/JSON) with provenance (grid, BCs, solver version/hash, seed).
- REGION: a mask or list of PASS points defining an existence region $\mathcal{R}$ with at least one PASS voxel.
- REPORT: compact JSON summarizing verdict (natural/maintained/fail), thresholds, maintenance cost ΔM , robustness τ (with CI), and hashes.

JSON policy: sanitize NaN/Inf to null; enforce schema_version='1.0.0'; validate against published schemas; hash everything (SHA-256) with a manifest including a self-hash.

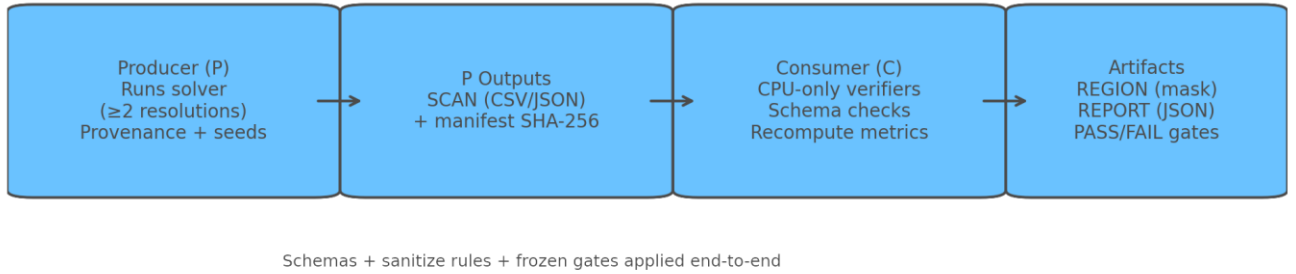


Figure 1. Verification pipeline: Producer (P) generates SCAN artifacts; Consumer (C) re-computes metrics, validates schemas/hashes, and emits REGION + REPORT under frozen gates. All PASS/FAIL results must be reproducible from SCAN alone (single source of truth).

3.4 Calibration (C) vs prediction (P) discipline

To limit degrees of freedom, SFT uses a single-pass calibration pipeline. Calibration outputs (q^* , $\hbar^*$, ϵ^* , μ^* , etc.) are frozen and reused across demonstrations without per-observable retuning. Any “ α -out” estimation is a separate, pre-registered exercise and does not affect the validity of α -in runs.

- α -in: α is treated strictly as an input used during EM calibration.
- α -out: optional pipeline estimating α from SFT-native observables without injecting α during calibration.
- Doc 10.1 (Alpha-Elasticity CI Suite) provides a CI/synthetic, verification-first protocol for alpha-like estimators (anti-leakage + preregistered gates) and should not be conflated with α -out.
- For limitations, preregistered acceptance bands, and the reviewer checklist (including External Break Test #01 as an anti-post-hoc audit), see Doc 8 — Limitations & Validation Plan.

4. Verification protocol and gates

This section summarizes default gates used in the distributed runner packs. Gates may be tightened, but should never be loosened post hoc for a claim. All gates must be declared in the REPORT and repeated in the REGION/SCAN metadata. For any REAL run, all thresholds must be locked (pre-registered) before solver execution and referenced by `gate_id` in the REPORT.

See Table 3. Detailed module gates and current evidential status

Module / claim	Status	Key P output	C verifier	Metric	Default gate	Notes (CI vs REAL)
Energy/continuity QA	REAL-ready	ENERGY_REPORT.json	verify_energy.py	$\Delta E/E$	$\Delta E/E \leq 1e-3$	[REAL] Also report ledger/continuity residuals + convergence.
Dispersion (small-k)	REAL-ready	dispersion_smallk.csv	fit_dispersion.py	c, R^2_{smallk}	$R^2 \geq 0.98$; $ \Delta c /c \leq 1\%$	[REAL] Declare k-window; run ≥ 2 grids.
EOC / convergence (CI)	CI	EOC_synthetic.csv	test_eoc_ci.py	p, R^2	$1.70 \leq p \leq 2.05$; $R^2 \geq 0.995$	[CI] Pipeline health-check only (synthetic).
AtomX existence	CI	synthetic_atom_scan.csv	verify_atomX.py	split, compat, R^2	split ≤ 0.19 meV; compat ≤ 0.02 ; $R^2 \geq 0.98$	[CI] Demonstration thresholds; use ablation for maintained vs natural.
Hydrogen 2s–2p	CI	COMPATIBILITY_SCAN_H.*	verify_hydrogen.py	split, compat, R^2	split ≤ 0.10 meV; compat ≤ 0.02 ; $R^2 \geq 0.98$	[CI] Demonstration; REAL validation requires solver outputs + convergence.
Double slit	CI	visibility.csv, shift.csv	verify_double_slit.py	phase R^2 , V monotone	$R^2 \geq 0.98$; sat ≤ 0.10 ; monotone $\downarrow$	[CI] Synthetic by default; REAL requires solver-generated CSVs.
Venus perihelion (overlay)	REAL-ready	varpi_series.csv	verify_venus_real.py	slope, R^2	$ \text{slope} - 8.624984 \leq 0.20$; $R^2 \geq 0.999$	[REAL] Fit interval predeclared; report sensitivity.
α -out (optional)	REAL-ready	DISPERSION_LI_NEAR.json, HBAR_FROM_ROTATOR.json, Q_EPS_SOLUTION.json	compose_alpha_out.py	tier error	Tier-1 $\leq 1\%$; Tier-2 $\leq 1e-4$; Tier-3 $\leq 1e-6$	[CI+REAL] CI validates plumbing; tiers require REAL components without α injection.
PPN bridge (optional)	REAL-ready	ppn_summary.json	verify_ppn.py		$ \beta - 1 \leq 1e-2$; $ \gamma - 1 \leq 1e-2$ (placeholder)	[REAL] Thresholds must be pre-registered (locked in a gates file/commit) before any REAL run; REPORT must reference the gate_id.

Firedrake spin bend FR	Exploratory P-bridge	spin_bend_scan.csv + SFT_SPIN_BEND_SIGNATURE_SCAN_REPORT.json	included report verifier	FR 2π phase + director closure	27/27 PASS; max distance to $\pi/\pi \approx 3.54e-15$; max closure $\approx 3.71e-17$	[Exploratory] Robust topological signature; not a full electron derivation.
Mercury perihelion Firedrake 1PN	Exploratory P-bridge	varpi_series_mercury.csv + SFT_SOLAR_PERIHELION_FIREDRAKE_REPORT.json	included geometry gate	slope, R^2	$ \text{slope} - 42.980204 \leq 0.20$; $R^2 \geq 0.995$; observed slope ≈ 43.057807 , $R^2 \approx 1.0$	[Exploratory] Effective solar-system geometry gate; not a derivation of GR.
Venus perihelion Firedrake 1PN	Exploratory P-bridge	varpi_series_venus.csv + SFT_VENUS_FIREDRAKE_GEODESIC_REPORT.json	included geometry gate	slope, R^2	$ \text{slope} - 8.624984 \leq 0.20$; $R^2 \geq 0.995$; observed slope ≈ 8.684022 , $R^2 \approx 1.0$	[Exploratory] Same bridge logic as Mercury.
Firedrake double-slit	Exploratory mixed	visibility.csv, shift.csv	included report verifier	phase R^2 , visibility saturation	phase PASS; visibility saturation FAIL under sat ≤ 0.10	[Exploratory] Useful negative/hold result; not a headline validation result.
Firedrake Alpha-Out bridge	Incomplete by design	SFT_ALPHA_OUT_BRIDGE_REPORT.json	compose bridge	available c + missing components	status INCOMPLETE; missing $\hbar^*$, q^* , ϵ^*	[Roadmap] Prevents alpha-out overclaim until real components exist.

Note: An additional CI-only protocol for an **alpha-like estimator** (anti-leakage, preregistered decision rule hashing, null/adversarial controls) is provided in **Doc 10.1**. This is a verification methodology appendix and is not a physical α validation.

Important: CI bundles certify the verification pipeline only; physical claims require REAL solver outputs under the published gates.

Synthetic CI modules certify tooling integrity (schemas, manifests, regression gates) in CPU-only environments. Scientific claims require P outputs produced by a real solver on at least two resolutions under the declared gates.

5. Alpha as a normalized elastic property of the structural medium

We adopt a monistic structural-medium view: all observables arise as stable configurations of a single field S . In this context, dimensionless couplings may be interpreted as normalized material properties of the underlying medium rather than arbitrary numbers.

We define a measurable stiffness proxy via the energetic response to controlled perturbations. For a baseline state S_0 , impose $\delta S(x) = \epsilon \cos(k \cdot x)$ and compute the energy increment $\Delta E(\epsilon, k)$. The effective spectral stiffness is $K_{\text{eff}}(k) = (1/V) * \partial^2 \Delta E / \partial \epsilon^2 \big|_{\epsilon=0}$. This definition is operational and directly auditable from solver outputs.

In SFT the fine-structure constant is expressed as $\alpha = q^2 / (4\pi \epsilon^* \hbar^* c)$, highlighting α as a dimensionless ratio between an emergent electrostatic scale ($q^2 / 4\pi \epsilon^*$) and the medium's action-propagation scale ($\hbar^* c$). We therefore interpret α as a normalized compliance parameter of the structural medium. An extended note (definitions, dimensional sanity, and external-verification contract) is provided in Appendix C.

$$K_{\text{eff}}(k) = (1/V) * \partial^2 \Delta E / \partial \epsilon^2 \big|_{\epsilon=0}$$

$$\alpha = q^2 / (4\pi \epsilon^* \hbar^* c)$$

6. External verification contract

6.1 Producer (P): what third parties must generate

To evaluate SFT claims, a third party must generate solver outputs with full provenance and at least two resolutions:

- Solver provenance: version/hash, grid(s), $\Delta t/\text{CFL}$, BCs, seeds, and any AMR thresholds.
- Integrity: MANIFEST_SHA256.txt with SHA-256 of all exported files (plus self-hash) [3].

6.2 Consumer (C): CPU-only verification

- Verify hashes and schema compliance (schema_version='1.0.0', JSON sanitize) [3-5].
- Recompute metrics and PASS masks from the SCAN artifacts (single source of truth).
- Emit REGION and REPORT artifacts with explicit thresholds and PASS/FAIL outcomes.
- An external Firedrake/UFL bridge mini-pack is also included; see Appendix A.

7. Firedrake/UFL bridge runs: selected audit targets (v0.1.9)

Parameter thread across Firedrake modules. In v0.1.9, all bridge modules share a common structural logic: each asks whether a declared configuration is compatible with the structural field/medium represented by S under its stated domain conditions. Local medium and discretization parameters are frozen per module prior to gate evaluation and are not retuned post hoc. The connecting thread is not a universal numerical parameter card, but the compatibility operator $\mathcal{C}[S;M]$ itself, applied consistently across spin, radial, perihelion, localization, and phase/coherence targets. All bridge results reported below come from the frozen package `SFT_FIREDRAKE_CORE_REPRO_v0_1_9` (SHA-256: 8740ABAC083450AFB5B2AF5AEB6A1AE634772E65D5088F27DFE0F80F36B23F83), executed in the same Firedrake/UFL and PETSc environment with `PETSC_DIR/PETSC_ARCH` unset and `OMP_NUM_THREADS=1`. The package does not impose one universal physical parameter value across heterogeneous tests; instead, each module carries its own declared medium/discretization block in its JSON report (mesh, polynomial degree, boundary conditions, thresholds, and body-specific constants where applicable). The common constraint is protocol-level: no per-observable retuning after gates, `schema_version='1.0.0'`, SHA-256 manifests, and machine-checkable PASS/FAIL gates. This is the parameter thread connecting the spin, radial, alpha-localization, double-slit, and perihelion bridge outputs.

This section records the first external-PDE/FEM bridge package for SFT-facing audit targets. The package is not treated as a global validation of SFT. Its narrower contribution is to show that selected medium-field diagnostics can be generated in a Firedrake/UFL environment and exported as auditable JSON/CSV artifacts with explicit PASS/FAIL outcomes.

7.1 Package status and integrity

The package ``SFT_FIREDRAKE_CORE_REPRO_v0_1_9`` contains real-solver/exploratory Firedrake bridge outputs for static and dynamic scalar-field diagnostics, spin FR signatures, radial Coulomb sanity checks, double-slit phase/visibility diagnostics, alpha-localization capacity scans, and solar-system perihelion geometry gates for Venus and Mercury. Package-level manifests and per-module manifests are included for audit.

Scalar Control And Topological Memory: Within this testbed, the result should be read as the combination of a negative scalar control and a positive topological-memory gate. The scalar recovery/residue runs disperse localized impulses without crossing the persistent-residue threshold, reducing the likelihood that the observed signal is caused by remnant scalar contamination. By contrast, the director/spin bridge retains a robust Finkelstein–Rubinstein memory across mesh resolution, loop resolution, and localization-width scans. Thus, within the stated model, artifact contract, and gates, the retained memory is best interpreted as topological/holonomic rather than as a scalar-residue artifact.

Interpretation rule. A Firedrake bridge PASS means that the exported artifact passed its declared gate inside this exploratory external solver environment. It does not by itself establish end-to-end SFT physical validation, nor does it remove the requirement for independent third-party runs at multiple resolutions.

7.2 Headline bridge results

Module	Artifact/report	Observed result	Gate status	Claim boundary
Scalar recovery/residue + spin topological memory	scalar recovery/residue reports + spin_topological_memory report	localized scalar impulses disperse without crossing the persistent-residue threshold; FR memory retained across mesh, loop-resolution, and localization-width scans	GLOBAL PASS	Negative scalar control + positive topological-memory gate; not director scattering dynamics.
Spin bend FR robustness	SFT_SPIN_BEND_SIGNATURE_SCAN_REPORT.json	27/27 cases PASS; max distance to $\pi/\pi \approx 3.54e-15$; max director closure $\approx 3.71e-17$	PASS	Robust spin-1/2 FR bend signature; not a full electron derivation.
Radial Coulomb H/T 2s-2p	SFT_RADIAL_COULOMB_FIREDRAKE_REPORT.json	H and T 2s/2p degeneracy PASS; max split approx $1.24e-5$ meV; max absolute energy error approx $1.43e-5$ meV under 1.0 meV gate	PASS	Hydrogenic radial boundary/topology sanity gate; not an SFT derivation of atoms.
Mercury perihelion 1PN	SFT_SOLAR_PERIHELION_FIREDRAKE_REPORT.json	slope approx 43.057807 arcsec/century vs 42.980204 expected; R2 approx 1.0	PASS	Solar geometry gate; not a derivation of GR or SFT gravity.
Venus perihelion 1PN	SFT_VENUS_FIREDRAKE_GEODESIC_REPORT.json	slope ≈ 8.684022 arcsec/century vs 8.624984 expected; $R^2 \approx 1.0$	PASS	Same geometry gate under Venus parameters.
Alpha localization capacity scan	SFT_ALPHA_LOCALIZATION_SCAN_REPORT.json	alpha_crit bracket [1.55, 1.6]; monotonic severity criteria PASS	PASS	Medium-capacity localization scan; not particle identification.

Double-slit Firedrake	SFT_DOUBLE_SLIT_FIRE DRAKE_REPORT.json	phase $R^2 = 1.0$ PASS; visibility saturation FAIL (last-step rel. change $\approx 0.274 > 0.10$)	MIXED / HOLD	Useful negative/hold result; keep separate from synthetic CI PASS.
Alpha-Out bridge	SFT_ALPHA_OUT_BRIDGE E_REPORT.json	status INCOMPLETE; missing $\hbar^*$, q^* , ϵ^*	INCOMPLETE	Roadmap artifact; prevents false alpha-out claim.

7.3 Recommended interpretation

The supported publication claim is: the verification-first contract has now been exercised by a Firedrake/UFL bridge on selected modules. The strongest current bridge results are the spin bend FR robustness scan, the radial Coulomb H/T 2s/2p sanity gate, and the two-body solar-system perihelion geometry gates (Venus and Mercury). The double-slit Firedrake branch should be reported as an honest mixed result: phase transport passes, while the visibility saturation gate currently fails under the declared threshold. Alpha-Out remains intentionally incomplete until $\hbar^*$, q^* , and ϵ^* are generated from SFT-native Firedrake artifacts.

7.4 Implications for the roadmap

The immediate next milestones are: (I) repeat headline bridge modules at at least two resolutions with locked gates; (II) produce a minimal GPU/Firedrake recipe for independent replication; (III) generate real alpha-out components (dispersion.csv, HBAR_FROM_ROTOR.json, coulomb_profile.csv, and field_energy.json) without alpha injection; and (iv) keep all FAIL or HOLD branches visible rather than tuning thresholds post hoc.

8. Limitations and roadmap (explicit)

We adopt strict scope and labeling to avoid overclaim.

- Experimental validation is pending for several proposed tests; this release focuses on making validation short and auditable.

- Scope is currently limited to EM + gravity mapping and an emergent spin- $\frac{1}{2}$ route; weak/flavor/hadronic completeness remains future work.
- Discrete vs continuum tension is treated as controlled systematics: $O((a k)^2)$ corrections are measured and reported via dispersion and EOC scans.
- α -out is optional and must be reported separately from α -in calibration; it is a falsifiable add-on via predeclared tiers.

Roadmap (verification-first):

1. Complete a Global Consistency Index (GCI) across all validated systems under a single frozen parameter set.
2. Add tests/verifiers for EM mapping uniqueness/domain (fixed operator + gauge fixing; no extra DOF; BC sensitivity).
3. Replicate the Venus/Mercury Firedrake perihelion gates independently at ≥ 2 resolutions with locked gates and published manifests.
4. Publish a minimal GPU run recipe for third parties (config + expected output files + gates) per module.
5. Quantum collapse (readout) is treated as an effective rule in this release.

8.1 Known failure modes (reviewer checklist)

To keep the evaluation robust and non-permissive, reviewers should treat the following as common failure modes and reject runs that exhibit them unless explicitly pre-registered and justified.

- Leakage (features vs targets): any predictor derived using the target observable (directly or via rescaling) invalidates the claim. Use strict C vs P discipline and group-wise LOO where applicable.
- Post-hoc gate changes: thresholds must be declared before solver runs. Tightening is allowed only if applied retroactively to all runs; loosening is disallowed.
- Insufficient resolution / pre-asymptotic regime: claims require ≥ 2 resolutions and evidence of convergence (or an explicit pre-asymptotic analysis).
- Boundary-condition artifacts: show BC sensitivity (or mitigation) and document BC choices in SCAN metadata; reject results that vanish under modest BC perturbations.
- Non-reproducible PASS masks: PASS/FAIL must be reconstructible from SCAN alone (single source of truth) using the published verifiers and schemas.

9. Conclusion

We released SFT as a verification-ready protocol: a minimal operational dictionary plus externally auditable artifacts (SCAN -> REGION -> REPORT), schemas, hash manifests, and preregistered pass/fail gates. This v0.3 draft adds a Firedrake/UFL bridge package with selected real-solver/exploratory audit targets. The strongest bridge results are the robust spin bend FR scan, the radial Coulomb H/T 2s/2p sanity gate, and the Venus/Mercury perihelion geometry gates; the double-slit Firedrake branch is deliberately reported as mixed, and Alpha-Out remains incomplete until real $\hbar^*$, q^* , and ϵ^* components are generated. The present status is therefore stronger than a pure CI protocol, but still not a global validation of SFT. Physical confirmation requires independent third-party REAL solver outputs at multiple resolutions under the published gates.

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Appendix A. Notation and conventions (short)

- Signature and units; normalization of S in the discrete model.
- Definition of c from small- k dispersion; definition of correlation length if used.
- PPN mapping conventions if the gravity bridge is tested.

Hashing standard: see Ref. [3].

Schema standard: see Ref. [5].

Bundles included (representative):

- External Review Pack [CI] (ALL_CHECKSUMS): Top-level reviewer bundle: schema validation + manifest checks + demo SCAN→REGION→REPORT artifacts.
- AtomX [CI] (Designed matter demo): Step-by-step recipe + CI scan/region/report; illustrates natural vs maintained via declared controls/ablation.
- Firedrake/UFL core bridge v0.1.9 [exploratory P-bridge]: a reference executable mini-pack implementing selected scalar-field equations, benchmarks, and exploratory diagnostics in a non-native PDE/FEM environment. Its purpose is not to reproduce the full SFT corpus externally, but to provide an auditable bridge for inspecting parts of the framework outside the native SFT runner ecosystem.
- Hydrogen radial [CI] (Phase-2 golden): Numerical QA bundle: 2s–2p degeneracy gate, normalization + virial checks (solver correctness sanity test).
- EOC CI synthetic [CI]: CPU-only regression test ensuring observed order-of-convergence tooling behaves correctly (synthetic smoke test).
- Optics null tests [CI]: Birefringence/dispersion null-analysis gates; demonstrates proceed/hold logic and stability checks.
- Double-slit verifier [CI]: Phase-fit and visibility saturation gates; synthetic-by-default CI harness + hooks for real solver CSVs.
- Venus perihelion verifier + real overlay [REAL-ready]: Extraction/fit/verification pipeline for perihelion slope; ready to consume third-party orbit outputs.
- Spin Appendix S runner [REAL-ready]: Fermi-rotation ($2\pi \rightarrow \pi$ phase) + rotor spectrum gates; robustness requires ≥ 2 grids and ≥ 2 seeds in REAL runs.
- α -out runner [REAL-ready] (with schema checks): Optional pipeline composing α from dispersion, rotor, and Coulomb/energy components; CI validates plumbing; REAL needed for tiers.

- Alpha-Elasticity CI Suite [CI] (Doc 10.1): Synthetic verification protocol for an alpha-like estimator (DECISION_RULE.json + decision_rule_sha256 + null/adversarial gates + manifest integrity). Demonstrates PASS/FAIL audit semantics; not a physical α claim.
- Firedrake/UFL core bridge v0.1.9 [exploratory P-bridge]: external PDE/FEM bridge with spin FR robustness scan, Venus/Mercury perihelion geometry gates, radial Coulomb sanity runs, alpha-localization capacity scan, double-slit phase/visibility diagnostics, and an intentionally incomplete Alpha-Out bridge report.

Quickstart (reviewer):

1) Verify hashes → 2) Validate schemas → 3) Run verifiers on provided artifacts → 4) Swap synthetic CI inputs for real solver outputs (P) and re-run gates.

Recommended minimal external (P) deliverables for a ‘REAL’ verification run:

- At least two resolutions (grids) per module; report convergence or GCI.
- Full provenance (solver version/hash, BCs, Δt /CFL, seed(s), platform).
- Module outputs exactly as named in Table 3 (or documented equivalents), plus MANIFEST_SHA256.txt.
- No post-hoc gate changes: thresholds must be declared before running the solver.

Appendix B. Alpha note (extended)

This appendix contains the longer, reviewer-safe note supporting Section 5. It is included for completeness and for external verifiers who wish to implement α -out and stiffness diagnostics.

Purpose. This note rewrites the intuition “alpha measures the rigidity of the S medium” in a reviewer-safe way: dimensionally consistent, operationally defined, and externally verifiable without requiring the author to run GPU-scale simulations.

1. Paper-ready text (3 paragraphs)

We adopt a monistic structural-medium view: all observables arise as stable configurations of a single field S . In this context, dimensionless couplings are naturally reinterpretable as normalized material properties of the underlying medium rather than as arbitrary numbers.

We define a measurable stiffness proxy of the medium through its energetic response to controlled perturbations. For a baseline state S_0 , impose $\delta S(x) = \epsilon \cos(k \cdot x)$ and compute the total energy increment $\Delta E(\epsilon, k)$. The effective spectral stiffness is defined as $K_{\text{eff}}(k) = (1/V) * \partial^2 \Delta E / \partial \epsilon^2 |_{\epsilon=0}$. This definition is operational and directly auditable from solver outputs.

In SFT the fine-structure constant is expressed as $\alpha = q^*{}^2 / (4\pi \epsilon^* \hbar^* c)$. This highlights α as a dimensionless ratio between an emergent electrostatic scale ($q^*{}^2/4\pi \epsilon^*$) and the structural action–propagation scale ($\hbar^* c$) of the medium. We therefore interpret α as a normalized compliance parameter of the structural medium; its quantitative value is delegated to external verification via the α -out pipeline.

2. Definitions and dimensional sanity

Key point: α is dimensionless. A Young’s modulus E has dimensions (energy/volume). So the correct statement is not “ α equals Young’s modulus”, but that α can parameterize a normalized stiffness/compliance once a reference scale is fixed by the model’s dimensional constants (a , t_0 , κ , $\hbar^*$, q^* , etc.).

3. Operational stiffness proxy (external-verification friendly)

Baseline: choose a background state S_0 (vacuum or specified steady background).

Perturbation: $\delta S(x) = \epsilon \cos(k \cdot x)$ with small ϵ and chosen wavevector k .

Measurement: for each (ϵ, k) , compute total energy $E(\epsilon, k)$ from the solver’s energy functional.

Definition: $\Delta E(\epsilon, k) = E(\epsilon, k) - E(0, k)$. Then define:

$K_{\text{eff}}(k) = (1/V) * (\partial^2 \Delta E / \partial \epsilon^2)$ evaluated at $\epsilon = 0$.

Practical estimation: evaluate ΔE at a small symmetric set of ϵ values (e.g., $\pm \epsilon_1, \pm \epsilon_2$) and fit a quadratic in ϵ . Report $K_{\text{eff}}(k)$ with confidence intervals and convergence checks across at least two grid resolutions.

4. Length scales without ad hoc assumptions

If the linearized dynamics around S_0 admits a dispersion $\omega^2 \approx c^2 k^2 + m_{\text{eff}}^2$, then a natural correlation length is:

$\xi = c / m_{\text{eff}}$.

Here m_{eff}^2 can be estimated either from the curvature of the effective potential around S_0 ($V''(S_0)$) or from a small- k dispersion fit. This avoids asserting $\xi = 1/\alpha$ without derivation.

5. What “alpha as rigidity” means in SFT (reviewer-safe wording)

Because $\alpha = q^2/(4\pi \epsilon^* \hbar^* c)$, alpha quantifies how strongly the emergent electrostatic scale competes with the medium’s action–propagation scale. Interpreting alpha as a normalized compliance is therefore a statement about a ratio of scales, not an identification with a dimensional modulus.

Optional (one-sentence intuition): smaller alpha corresponds to a medium that is harder to polarize into EM response per unit structural action–propagation, in the chosen normalization.

6. External verification contract (minimal P outputs + C verifiers)

To validate the interpretation with third-party compute, separate:

P (producer): generates solver outputs on target grids/BCs.

C (consumer): runs CPU-only verifiers that compute $K_{\text{eff}}(k)$, ξ , and alpha-out components.

Minimum producer outputs (examples):

energy_vs_eps_k.csv (or JSON) with $E(\epsilon, k)$ on at least two grids.

dispersion_smallk.csv with $\omega(k)$ for small k on matching grids.

coulomb_profile.csv and field_energy.json (if composing q^* , ϵ^*), using declared windows/cutoffs.

rotor_spectrum.csv plus $I(S)$ report (if composing $\hbar^*$).

Minimum consumer verifiers (CPU-only):

fit_stiffness.py: estimates $K_{\text{eff}}(k)$ + CI + convergence summary.

fit_dispersion.py: estimates c and m_{eff} (and thus ξ) + CI.

compose_alpha_out.py: composes alpha-out with tier thresholds.

7. Framing notes (avoid overclaim)

Use “interpret” / “parameterizes” / “is consistent with” rather than “equals” when speaking about elastic moduli. Mark toy profiles (e.g., hedgehog) as illustrations, not derivations. Reserve “validated” for cases where producer (P) outputs from the real solver are provided and pass the published gates.